Submission B solution to Problem 7

Overview / Strategy

The answer to the original question is negative. We construct an explicit reducible quasi-reduced word
$$w = a\,b\,B\,A\,c\,C\,a\,d\,D\,A\,e\,E\,a\,f\,F\,A$$
in the free group on a, b, c, d, e, f, where capital letters denote inverses, and prove that the associated complex F_w has nonzero $H_2(-;\mathbb{F}_2)$. The proof has three main moves. First, planar topology implies that the subdivided crossing graph of every admissible matching is a forest, and for a forest the admissibility condition is exactly a condition on cyclically consecutive leaves of each tree component. This is done in Lemmas 4 and 5.

Second, the five uniquely occurring inverse pairs bB, cC, dD, eE, fF are forced to be isolated uncrossed caps. Removing these caps leaves a six-point "skeleton" with labels a, A, a, A, a, A. In this skeleton we exhibit three admissible two-cells Q_1, Q_2, Q_3 whose mod-2 cellular boundary cancels. Finally, the forest condition gives a crossing bound showing that no admissible matching for this word has three or more crossings. Hence the constructed two-cycle cannot be a boundary, so $H_2(F_w;\mathbb{F}_2) \neq 0$.

1 Definitions and headline theorem

Let
$$G = \langle a_1, \ldots, a_r \rangle$$
be a free group, and let
$$L = \{a_1, a_1^{-1}, \ldots, a_r, a_r^{-1}\}.$$
A word $w = w_1 \cdots w_m$, with $w_i \in L$, is called *reducible* if its value in G is the identity. It is called *quasi-reduced* if it contains no consecutive subword of the form
$$\ell\,\ell^{-1}\,\ell$$
for any $\ell \in L$.

Throughout,
$$\mathbb{H} = \mathbb{R} \times [0, \infty), \qquad \mathbb{H}^\circ = \mathbb{R} \times (0, \infty).$$
For a word $w = w_1 \cdots w_{2n}$, a *matching* for w is a union M of n compact intervals, each properly embedded in $\mathbb{H}$, such that:

- the endpoints of the intervals are exactly

$$\partial M = \{1, \ldots, 2n\} \times \{0\};$$

- the interiors of the intervals lie in $\mathbb{H}^\circ$, any two intervals meet only in finitely many transverse double points, and there are no triple points;
- for each component R of $\mathbb{H} \setminus M$, and for each component A of $M \cap \partial \overline{R}$, the boundary ∂A consists of two marked boundary points $(i, 0)$ and $(j, 0)$ with

$$w_i = w_j^{-1}.$$

A matching is a *k-crossing matching* if the total number of transverse double points is k.

The CW complex F_w is the following complex. Its k-cells are indexed by isotopy classes, relative to the marked boundary points, of k-crossing matchings. If M is such a matching and $X(M)$ is its crossing set, the corresponding closed cell is

$$C(M) = \prod_{c \in X(M)} I_c,$$

where I_c is an interval whose two endpoints correspond to the two local resolutions of the crossing c. If a facet is obtained by fixing the coordinate I_c at an endpoint $a \in \partial I_c$, then the facet is attached to the cell corresponding to the matching M_a obtained from M by resolving c according to a. The remaining coordinates are identified canonically with the crossings of M_a.

Theorem 1 (Headline theorem: the original question has a negative answer)**.** *There exists a reducible quasi-reduced word w for which the associated complex F_w is not contractible. More precisely, let*

$$G = \langle a, b, c, d, e, f \rangle,$$

write

$$A = a^{-1}, \quad B = b^{-1}, \quad C = c^{-1}, \quad D = d^{-1}, \quad E = e^{-1}, \quad F = f^{-1},$$

and set

$$w = a\, b\, B\, A\, c\, C\, a\, d\, D\, A\, e\, E\, a\, f\, F\, A.$$

Then w is reducible and quasi-reduced, but

$$H_2(F_w; \mathbb{F}_2) \neq 0.$$

Consequently F_w is not contractible.

The proof of Theorem 1 is completed at the end, after the necessary planar and cellular arguments have been established.

2 The word

Proposition 2. *The word*

$$w = a\,b\,B\,A\,c\,C\,a\,d\,D\,A\,e\,E\,a\,f\,F\,A$$

is reducible and quasi-reduced.

Proof. The word freely reduces to the identity. Indeed, cancelling the adjacent inverse pairs

$$bB, \qquad cC, \qquad dD, \qquad eE, \qquad fF$$

leaves

$$aA\,aA\,aA,$$

which freely reduces to 1. Thus $w = 1 \in G$.

It remains to check that w is quasi-reduced. A forbidden subword $\ell\ell^{-1}\ell$ must begin with an adjacent inverse pair. The only adjacent inverse pairs in the displayed word, in either orientation, are

$$bB, \qquad cC, \qquad dD, \qquad eE, \qquad fF.$$

The letters immediately following these pairs are respectively

$$A, \qquad a, \qquad A, \qquad a, \qquad A.$$

These are not respectively

$$b, \qquad c, \qquad d, \qquad e, \qquad f.$$

Therefore no adjacent inverse pair begins a subword of the form $\ell\ell^{-1}\ell$. Hence w is quasi-reduced. □

3 Planar structure of admissible matchings

We use the Jordan curve theorem and Euler's formula for finite planar graphs in their standard forms; see, for example, [2].

Definition 3. Let M be a matching. Its *subdivided crossing graph* $\Gamma(M)$ is the finite graph whose vertices are the boundary endpoints of the intervals together with the crossing points, and whose edges are the arc segments of the intervals between consecutive vertices. Boundary endpoints have degree 1, and crossing vertices have degree 4.

Lemma 4. *For every admissible matching M, the graph $\Gamma(M)$ is a forest.*

Proof. Suppose, toward a contradiction, that $\Gamma(M)$ contains a graph cycle γ. Since boundary endpoints have degree 1, the cycle cannot pass through a boundary endpoint. Hence γ lies in $\mathbb{H}^\circ$. Its image is a simple closed curve. By the Jordan curve theorem, it bounds a disk Δ. The disk Δ is contained in $\mathbb{H}^\circ$: any point with $y \leq 0$ can be joined to infinity through the closed lower half-plane without meeting γ, so it lies in the unbounded component of the complement of γ.

Since M is a finite union of arcs, $\Delta \setminus M$ is nonempty. Let U be a component of $\Delta \setminus M$, and let R be the component of $\mathbb{H} \setminus M$ containing U. Because $\partial\Delta = \gamma \subset M$, a path in R cannot cross from U to the exterior of Δ. Thus $R \subset \Delta$. In particular $\partial\overline{R}$ is disjoint from $\mathbb{R} \times \{0\}$.

The set $M \cap \partial\overline{R}$ is nonempty, since R is a bounded complementary component of a nonempty finite graph. But every component of $M \cap \partial\overline{R}$ is disjoint from the boundary line $\mathbb{R} \times \{0\}$, and therefore no such component can have boundary equal to two marked boundary points $(i, 0)$ and $(j, 0)$. This contradicts admissibility. Hence $\Gamma(M)$ has no cycles, so it is a forest. $\square$

Lemma 5 (Side paths in a tree component)**.** *Let M be a matching such that $\Gamma(M)$ is a forest. Let T be a tree component of $\Gamma(M)$, and let its boundary leaves be*

$$x_1 < x_2 < \cdots < x_m$$

in their order along $\mathbb{R} \times \{0\}$, with x_m and x_1 regarded as cyclically consecutive through infinity. For $i = 1, \ldots, m$, with $x_{m+1} = x_1$, let P_i be the unique path in T from x_i to x_{i+1}.

As R ranges over the components of $\mathbb{H} \setminus M$, the components of $M \cap \partial\overline{R}$ lying in T are exactly the paths

$$P_1, \ldots, P_m.$$

Consequently, for a forest matching, the admissibility label condition is equivalent to the following condition: for every tree component T, cyclically consecutive boundary leaves of T have inverse labels.

Proof. First consider only the tree T. Compactify $\mathbb{H}$ by adding the point at infinity to the boundary, obtaining a closed disk D. The boundary leaves $x_1, \ldots, x_m$ lie on ∂D. Let α_i be the boundary arc from x_i to x_{i+1}, with indices taken cyclically.

Form the planar graph

$$K = T \cup \partial D.$$

If T has v vertices, then T has $v - 1$ edges. The boundary circle ∂D is cut by the m leaves into m boundary edges. Thus

$$V(K) = v, \qquad E(K) = v - 1 + m.$$

Euler's formula on the sphere gives

$$F(K) = 2 - V(K) + E(K) = 2 - v + (v - 1 + m) = m + 1.$$

One face is the exterior of the disk D, so $D \setminus T$ has exactly m components.

No component of $D \setminus T$ is disjoint from ∂D. Indeed, an interior face whose boundary did not meet ∂D would have a boundary component containing a simple closed curve made of edges of T, contradicting that T is a tree. Hence every component of $D \setminus T$ contains at least one of the arcs α_i. Since there are m components and m such arcs, each component contains exactly one α_i. Let U_i be the component containing α_i.

We now show that the part of $\partial\overline{U_i}$ lying in T is exactly P_i. The set

$$J_i = P_i \cup \alpha_i$$

is a simple closed curve. Let $\Delta_i \subset D$ be the disk bounded by J_i on the side adjacent to the boundary arc α_i, so that

$$\Delta_i \cap \partial D = \alpha_i.$$

We claim that

$$T \cap \Delta_i = P_i.$$

If not, some component S of $T \setminus P_i$ meets the interior of Δ_i. Since T is a tree, S attaches to P_i at exactly one vertex; two distinct attachment vertices would give a cycle in T. The finite tree S has a leaf not equal to its attachment point, and because all degree-one vertices of T are boundary leaves, this leaf is some boundary leaf $y \neq x_i, x_{i+1}$. The path in S from the attachment point to y starts inside Δ_i. To end at y, which is not on $\alpha_i = \Delta_i \cap \partial D$, it would have to cross the Jordan curve J_i again. It cannot cross α_i, because T meets ∂D only at its leaves, and it cannot meet P_i again without creating a cycle in T. This is impossible. Therefore $T \cap \Delta_i = P_i$, and the T-part of $\partial\overline{U_i}$ is exactly P_i. This proves the lemma when the matching consists only of T.

Now restore the other tree components of M. They are compact and disjoint from T, hence disjoint from each path P_i. A sufficiently small one-sided collar of P_i inside the component U_i therefore avoids all other components of M. This collar lies in a single component R_i of $\mathbb{H} \setminus M$, and P_i is a component of $M \cap \partial\overline{R_i}$. Conversely, if a component of $M \cap \partial\overline{R}$ lies in T, then near any interior point of one of its edges the region R lies on one side of T, hence inside one of the components U_i of $D \setminus T$. The preceding paragraph shows that the corresponding side of T is exactly P_i. Thus the side paths of T in the full matching are precisely $P_1, \dots, P_m$.

The final equivalence follows directly. If the matching is admissible, then every side path P_i has endpoints with inverse labels, so cyclically consecutive leaves in each tree component have inverse labels. Conversely, if cyclically consecutive leaves in every tree component have inverse labels, then every component of $M \cap \partial\overline{R}$ is one of the paths P_i, and its endpoints satisfy the required inverse-label condition. Thus the admissibility condition is exactly the stated cyclic leaf condition. □

Lemma 6 (Unique inverse pairs are forced)**.** *Suppose a letter x and its inverse x^{-1} each occur exactly once in a word, at boundary points p and q. Then in every admissible matching, p and q are joined by an isolated uncrossed interval.*

Proof. Let M be an admissible matching, and let T be the tree component of $\Gamma(M)$ containing p. By Lemma 5, the two side paths of T incident to p end at the predecessor and successor of p among the boundary leaves of T, in cyclic order. Admissibility forces both of those endpoint labels to be x^{-1}. Since q is the unique boundary point with label x^{-1}, both the predecessor and the successor must be q. This is possible only if T has exactly two boundary leaves, namely p and q.

Let m be the number of boundary leaves of T, and let t be the number of crossing vertices of T. Since T is a tree,

$$E = (m + t) - 1.$$

On the other hand, boundary leaves have degree 1 and crossing vertices have degree 4, so the degree sum gives

$$2E = m + 4t.$$

Combining these equations,

$$2(m + t - 1) = m + 4t,$$

hence

$$m = 2t + 2.$$

Here $m = 2$, so $t = 0$. Thus T has no crossing vertices. A connected component with two boundary leaves and no crossing vertices is a single embedded interval joining those leaves. Since it is a whole component of $\Gamma(M)$, it is isolated from every other interval. Therefore p and q are joined by an isolated uncrossed interval. $\square$

Proposition 7 (Forced caps and the six-point skeleton)**.** *For the word*

$$w = a\, b\, B\, A\, c\, C\, a\, d\, D\, A\, e\, E\, a\, f\, F\, A,$$

every admissible matching contains the five isolated uncrossed intervals

$$(2, 3), \qquad (5, 6), \qquad (8, 9), \qquad (11, 12), \qquad (14, 15)$$

in the original numbering. Deleting these five intervals leaves six boundary points, originally

$$1,\ 4,\ 7,\ 10,\ 13,\ 16,$$

which we relabel as $1, \ldots, 6$*. Their labels are*

$$a,\ A,\ a,\ A,\ a,\ A.$$

Proof. The inverse pairs

$$b, B; \qquad c, C; \qquad d, D; \qquad e, E; \qquad f, F$$

each occur exactly once in w. Lemma 6 therefore forces the corresponding boundary points to be joined by isolated uncrossed intervals. In the original numbering these are exactly

$$(2, 3), \qquad (5, 6), \qquad (8, 9), \qquad (11, 12), \qquad (14, 15).$$

Each of these pairs consists of adjacent boundary points. An uncrossed interval joining adjacent boundary points, together with the boundary segment between them, bounds a disk containing no marked boundary point. Because the interval is isolated, no other interval can cross it; and because the boundary segment contains no marked boundary point, no other interval can enter that disk without crossing the interval. Thus each forced interval may be represented, up to isotopy, by a small boundary-parallel cap.

After deleting these five caps, the remaining original boundary points are

$$1,\ 4,\ 7,\ 10,\ 13,\ 16.$$

Their labels are respectively

$$a,\ A,\ a,\ A,\ a,\ A.$$

Relabelling them $1, \dots, 6$ gives the claimed six-point skeleton. □

4 A crossing bound

Lemma 8 (Crossing count in a forest)**.** *Suppose a matching has r intervals, k crossings, and c connected components in its subdivided crossing graph. If that graph is a forest, then*

$$k = r - c.$$

In particular, if $r > 0$, then $k \leq r - 1$.

Proof. The graph has

$$V = 2r + k$$

vertices: $2r$ boundary leaves and k crossing vertices. If an interval has s crossings, it is subdivided into $s + 1$ edges. Summing over all intervals gives

$$E = r + 2k,$$

because each crossing lies on two intervals. Since the graph is a forest with c connected components,

$$E = V - c.$$

Thus

$$r + 2k = 2r + k - c,$$

so

$$k = r - c.$$

If $r > 0$, then $c \geq 1$, and therefore $k \leq r - 1$. □

5 The two-cells and their facets

From now on we work in the six-point skeleton of Proposition 7; the five forced caps are always understood to be reinserted afterward. For skeleton endpoints $i < j$, write (ij) for an interval joining i and j. We use standard chord drawings in the upper half-plane, so two intervals (i, k) and (j, ℓ), with $i < k$ and $j < \ell$, cross exactly when their endpoints strictly interleave.

Proposition 9 (Three admissible two-cells)**.** *Define skeleton matchings*

$$Q_1 = (13)(25)(46), \qquad Q_2 = (14)(26)(35), \qquad Q_3 = (15)(24)(36).$$

After reinserting the five forced caps, each Q_i is an admissible two-crossing matching for w. Hence Q_1, Q_2, Q_3 define distinct 2-cells of F_w.

Proof. The crossing pairs are

skeleton matching	crossing pairs
Q_1	$(13) \times (25),\ (25) \times (46)$
Q_2	$(14) \times (26),\ (14) \times (35)$
Q_3	$(15) \times (36),\ (24) \times (36).$

Thus each Q_i has exactly two crossings. In each case the crossing graph on the three skeleton intervals is connected: for Q_1 the middle interval (25) crosses the other two, for Q_2 the interval (14) crosses the other two, and for Q_3 the interval (36) crosses the other two.

For each Q_i, the subdivided skeleton graph has 6 boundary leaves and 2 crossing vertices, so

$$V = 8.$$

It has 3 original intervals and 2 crossings, hence

$$E = 3 + 2 \cdot 2 = 7.$$

Since the graph is connected and $E = V - 1$, it is a tree.

The leaves of this tree are the skeleton endpoints

$$1, 2, 3, 4, 5, 6$$

in cyclic order, with labels

$$a, A, a, A, a, A.$$

Cyclically consecutive leaves therefore have inverse labels. The five forced caps are separate two-leaf tree components, and their endpoint labels are also inverse. Lemma 5 now implies that, after reinserting the forced caps, each Q_i is admissible. The three matchings have different endpoint pairings, and an isotopy fixing the marked boundary points preserves endpoint pairings. Hence they define distinct 2-cells. $\square$

Proposition 10 (One-cells and facet table)**.** *Define six skeleton matchings*

$$P_1 = (12)(35)(46), \quad P_2 = (15)(23)(46), \quad P_3 = (13)(24)(56),$$
$$P_4 = (13)(26)(45), \quad P_5 = (16)(24)(35), \quad P_6 = (15)(26)(34).$$

After reinserting the forced caps, each P_i is an admissible one-crossing matching for w. Moreover, over $\mathbb{F}_2$, the cellular boundaries of the two-cells above are

$$\partial Q_1 = P_1 + P_2 + P_3 + P_4,$$

$$\partial Q_2 = P_1 + P_4 + P_5 + P_6,$$

$$\partial Q_3 = P_2 + P_3 + P_5 + P_6.$$

Proof. Each P_i has exactly one crossing. The following table records the isolated inverse-labelled interval and the leaves of the crossing component:

P_i	isolated interval	leaves of crossing component
P_1	(12)	$3, 4, 5, 6$
P_2	(23)	$1, 4, 5, 6$
P_3	(56)	$1, 2, 3, 4$
P_4	(45)	$1, 2, 3, 6$
P_5	(16)	$2, 3, 4, 5$
P_6	(34)	$1, 2, 5, 6.$

In every row, the isolated interval has inverse endpoint labels, and the four leaves of the crossing component have alternating labels a, A, a, A in cyclic order. The crossing component is a tree with four leaves and one crossing vertex. Therefore Lemma 5, together with the forced caps, shows that each P_i is admissible.

For a crossing between intervals (i, k) and (j, ℓ), where

$$i < j < k < \ell,$$

the two local resolutions are

$$(i, j)(k, \ell) \qquad \text{and} \qquad (i, \ell)(j, k).$$

Applying this rule to the crossings of Q_1, Q_2, Q_3 gives:

two-cell	crossing resolved	resulting one-cells
Q_1	$(13) \times (25)$	$P_1,\ P_2$
Q_1	$(25) \times (46)$	$P_3,\ P_4$
Q_2	$(14) \times (26)$	$P_1,\ P_5$
Q_2	$(14) \times (35)$	$P_4,\ P_6$
Q_3	$(15) \times (36)$	$P_3,\ P_5$
Q_3	$(24) \times (36)$	$P_2,\ P_6.$

For example, resolving $(13)\times(25)$ in Q_1 gives either

$$(12)(35)(46) = P_1$$

or

$$(15)(23)(46) = P_2.$$

The other entries follow by the same resolution rule.

In the cellular chain complex over $\mathbb{F}_2$, orientation signs disappear. Each facet of a square $C(Q_i)$ maps canonically and homeomorphically to the one-cell obtained by the corresponding resolution. Therefore the cellular boundaries are exactly the sums of the four one-cells appearing in the two rows for that Q_i, which are the three displayed formulas. $\square$

6 No three-cells and nonzero homology

Proposition 11. *The complex F_w has no cells of dimension 3 or higher.*

Proof. Let M be any admissible matching for w. By Proposition 7, M contains the five forced isolated uncrossed caps

$$(2,3), \qquad (5,6), \qquad (8,9), \qquad (11,12), \qquad (14,15)$$

in the original numbering. These caps have no crossings.

After deleting them, exactly three intervals remain. Let k be the number of crossings among these three intervals. By Lemma 4, $\Gamma(M)$ is a forest. Deleting isolated tree components preserves the property of being a forest, so the subdivided crossing graph of the remaining three intervals is also a forest. If this remaining forest has c connected components, Lemma 8 gives

$$k = 3 - c \leq 2.$$

Thus every admissible matching for w has at most two crossings. Since k-cells of F_w correspond to k-crossing matchings, F_w has no cells in dimensions 3 or higher. $\square$

Proposition 12. *For the word w above,*

$$H_2(F_w; \mathbb{F}_2) \neq 0.$$

Proof. Work in cellular chains over $\mathbb{F}_2$, and write the same symbol for a cell and for its corresponding cellular generator. By Proposition 10,

$$\partial Q_1 = P_1 + P_2 + P_3 + P_4,$$

$$\partial Q_2 = P_1 + P_4 + P_5 + P_6,$$

$$\partial Q_3 = P_2 + P_3 + P_5 + P_6.$$

Adding these three equations over $\mathbb{F}_2$, each P_i appears exactly twice. Hence

$$\partial(Q_1 + Q_2 + Q_3) = 0.$$

Thus

$$z \coloneqq Q_1 + Q_2 + Q_3$$

is a cellular 2-cycle over $\mathbb{F}_2$.

The chain z is nonzero. Indeed, cellular chain groups are free $\mathbb{F}_2$-vector spaces on the cells, and Q_1, Q_2, Q_3 are distinct 2-cells by Proposition 9. By Proposition 11, there are no 3-cells, so

$$C_3(F_w; \mathbb{F}_2) = 0.$$

Therefore

$$B_2(F_w; \mathbb{F}_2) = \operatorname{im}\big(\partial : C_3(F_w; \mathbb{F}_2) \to C_2(F_w; \mathbb{F}_2)\big) = 0.$$

Since $z \neq 0$ and z is a cycle, it represents a nonzero cellular homology class. By the cellular homology theorem [1, Chapter 2], cellular homology agrees with ordinary singular homology for CW complexes. Therefore

$$H_2(F_w; \mathbb{F}_2) \neq 0.$$

□

Proof of Theorem 1. Proposition 2 shows that the displayed word w is reducible and quasi-reduced. Proposition 12 shows that

$$H_2(F_w; \mathbb{F}_2) \neq 0.$$

A contractible space has trivial homology in every positive degree over every coefficient field. Hence F_w is not contractible. This gives a negative answer to the original question. □

References

[1] Allen Hatcher, *Algebraic Topology*, Cambridge University Press, 2002.

[2] James R. Munkres, *Topology*, Second edition, Prentice Hall, 2000.